PulseIQ Analytics | Phase 3

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Track: Data Analysts
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GitHub: PoonamPatole/pulseiq
Git tag: Phase-3

Task status and evidence

Task	Status	Evidence
11. P&L and expenses	Complete locally; P&L ties to governed M16-M18 views.	Findings and tie-out
12. Receivables and cash flow	Complete locally; ageing, payment timing, and cash measures built.	Bucket check
13. Cross-department impact	Complete locally; 13 shared measures reconcile to PQ-13.	Comparison notes
14. AI signals and forecasts	Complete locally; anomalies and forecasts include uncertainty notes.	Forecast notes
15. Executive scorecard	Complete locally; eight governed KPIs and supplied targets.	KPI selection

Proof it works

Task 11: Warehouse counts match the source: 6,784 ledger and 501 expense rows. Revenue, net profit, and margin tie to M16-M18 for April 2025-September 2026. From March to April 2026, revenue rose 11.4%, expenses 20.7%, and profit fell from INR 15.48m to INR 11.45m.

Task 12: Receivables use four non-overlapping age bands; their totals check against open balances. Cash-in and cash-out follow payment dates, with GST included for paid bills.

Task 13: All 2,574 checks (198 department-months x 13 measures) match PQ-13 after eight documented exclusions. Source records remain unchanged.

Task 14: 405 anomalies include INR 31.42m estimated payroll exposure across 184 records; this is for review, not confirmed loss. Twelve forecasts have no actuals yet, so error and MAPE are unavailable.

Task 15: The scorecard has eight governed KPIs. Targets are used only when supplied; four KPIs have no numeric target.

What I learned

I learned to keep financial measures tied to their source and date basis, aggregate facts to a shared grain before joining, and present model signals as uncertain. Targets should come from business owners, not inference.

Problems and how I handled them

The workbook headlines are counts, so Task 11 ties to governed metrics. Task 13 documents eight excluded records and preserves source facts. One approved leave request has reversed dates and is assigned to its start month for review. Task 14 has no actuals in its forecast horizon; blank anomaly estimates are not treated as zero. Task 15 leaves four KPIs without supplied targets unscored.

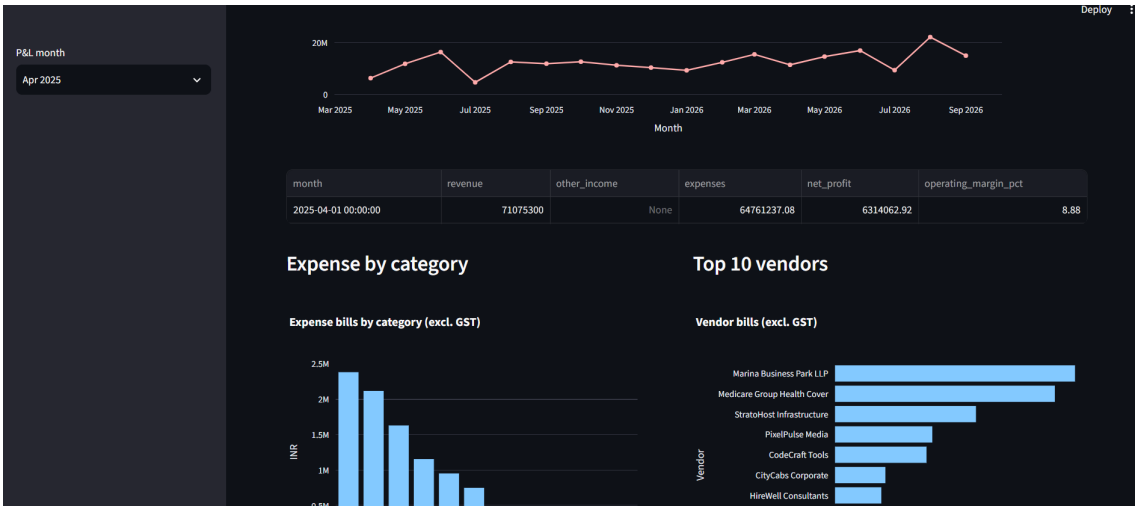
Plan for the next phase

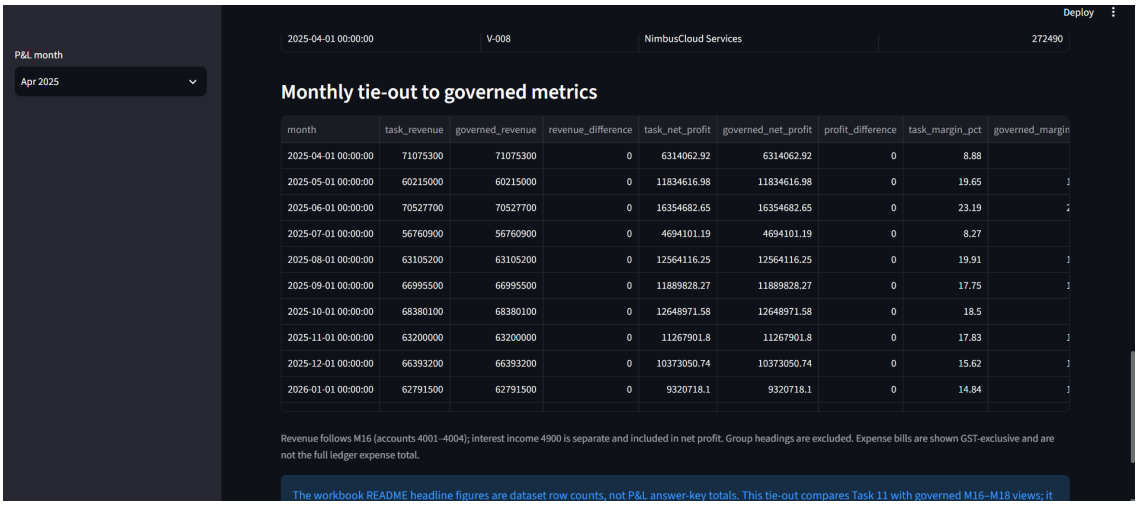
Review payroll and source exceptions with their owners, compare forecasts when actuals arrive, confirm KPI targets, and package the dashboards and evidence for final submission.

Task screenshots

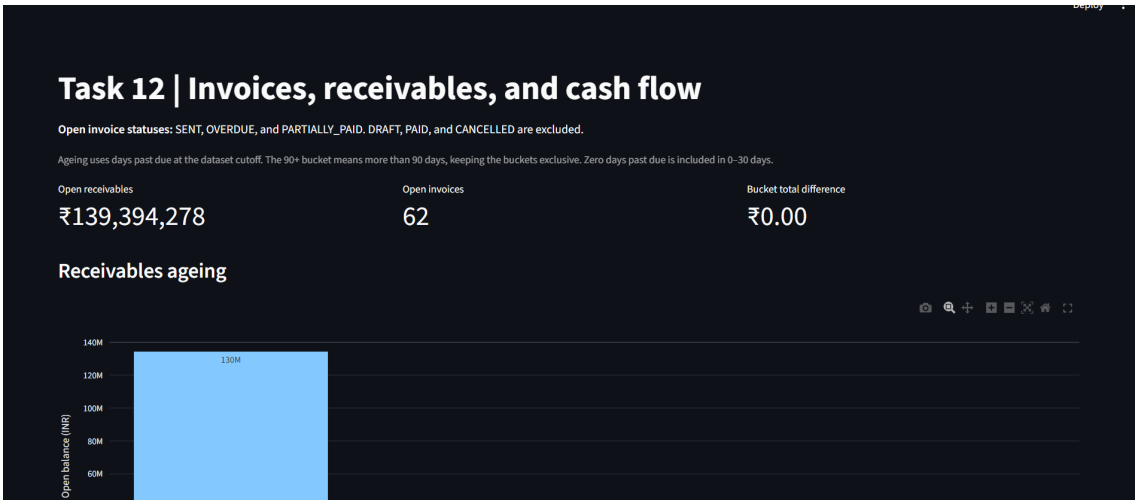
Task 11. P&L and expense analytics



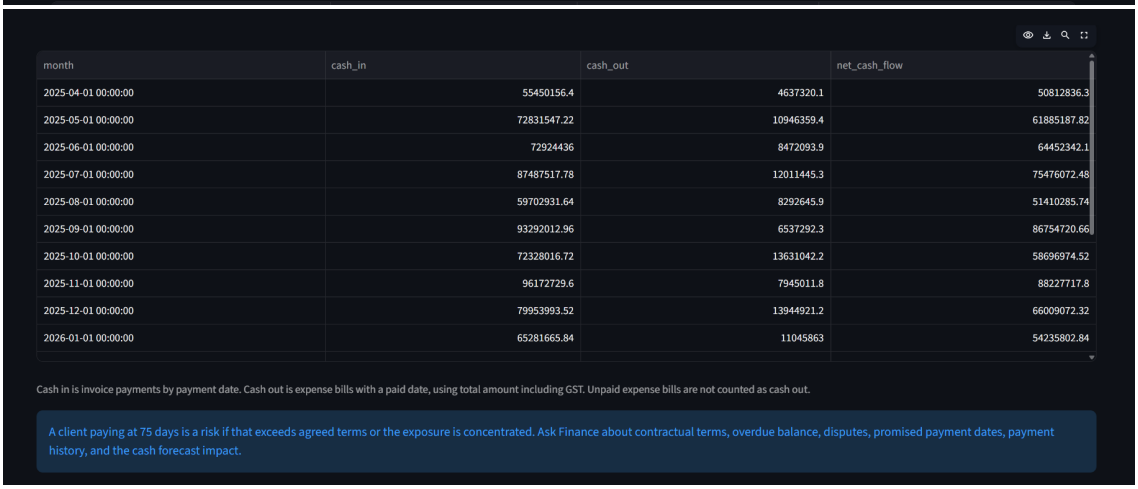




Task 12. Invoices, receivables, and cash flow





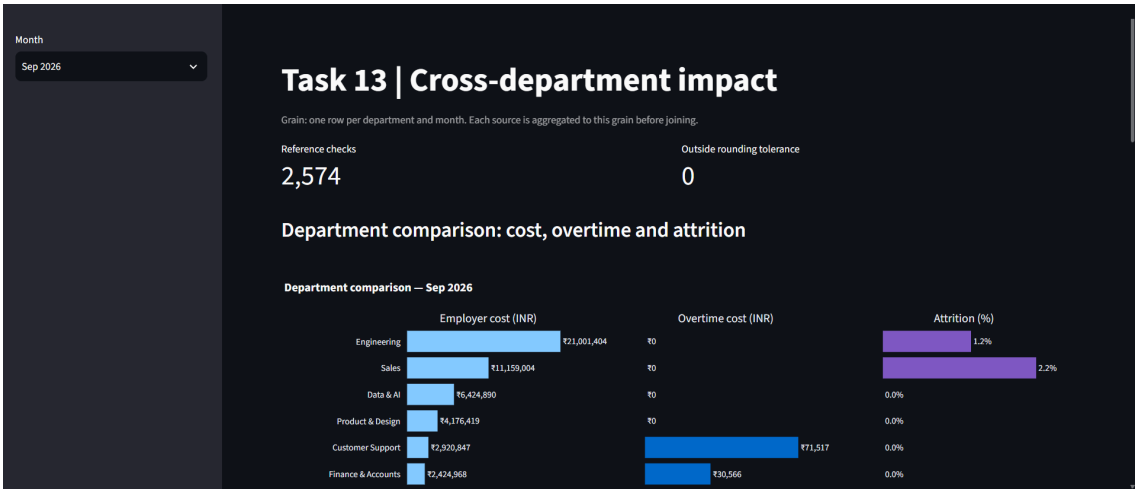


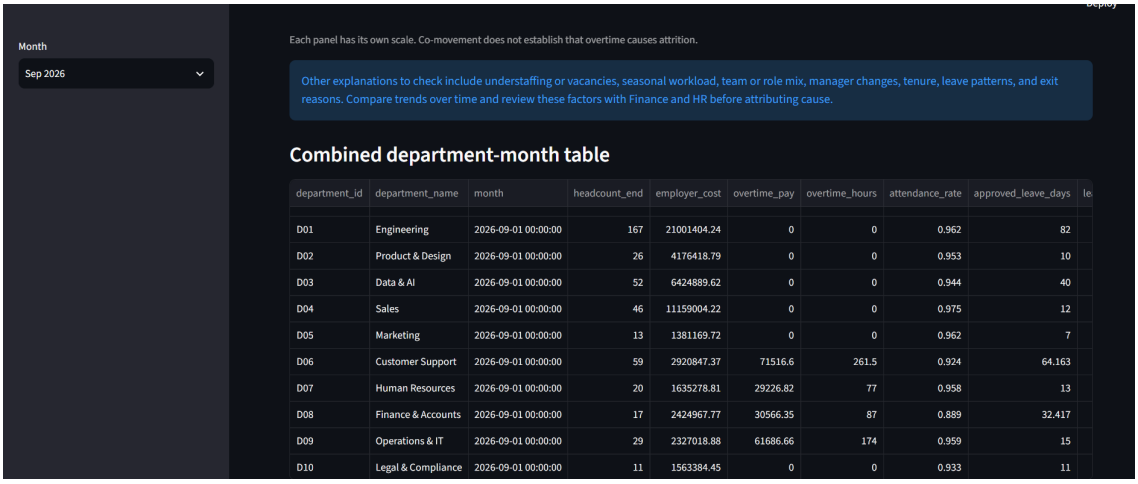
month	cash_in	cash_out	net_cash_flow
2025-04-01 00:00:00	55450156.4	4637320.1	50812836.3
2025-05-01 00:00:00	72831547.22	10946359.4	61885187.82
2025-06-01 00:00:00	72924436	8472093.9	64452342.1
2025-07-01 00:00:00	87487517.78	12011445.3	75476072.48
2025-08-01 00:00:00	59702931.64	8292645.9	51410285.74
2025-09-01 00:00:00	93292012.96	6537292.3	86754720.66
2025-10-01 00:00:00	72328016.72	13631042.2	58696974.52
2025-11-01 00:00:00	96172729.6	7945011.8	88227717.8
2025-12-01 00:00:00	79953993.52	13944921.2	66009072.32
2026-01-01 00:00:00	65281665.84	11045863	54235802.84

Cash in is invoice payments by payment date. Cash out is expense bills with a paid date, using total amount including GST. Unpaid expense bills are not counted as cash out.

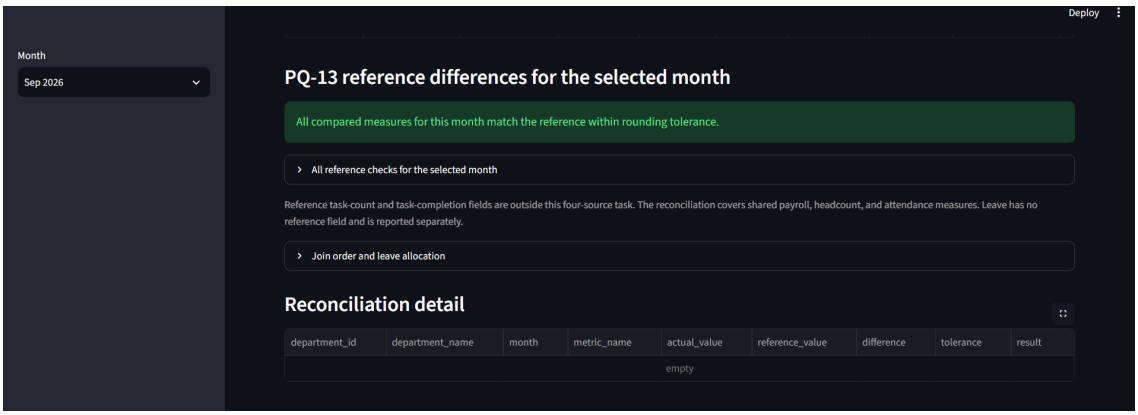
A client paying at 75 days is a risk if that exceeds agreed terms or the exposure is concentrated. Ask Finance about contractual terms, overdue balance, disputes, promised payment dates, payment history, and the cash forecast impact.

Task 13. Cross-department impact



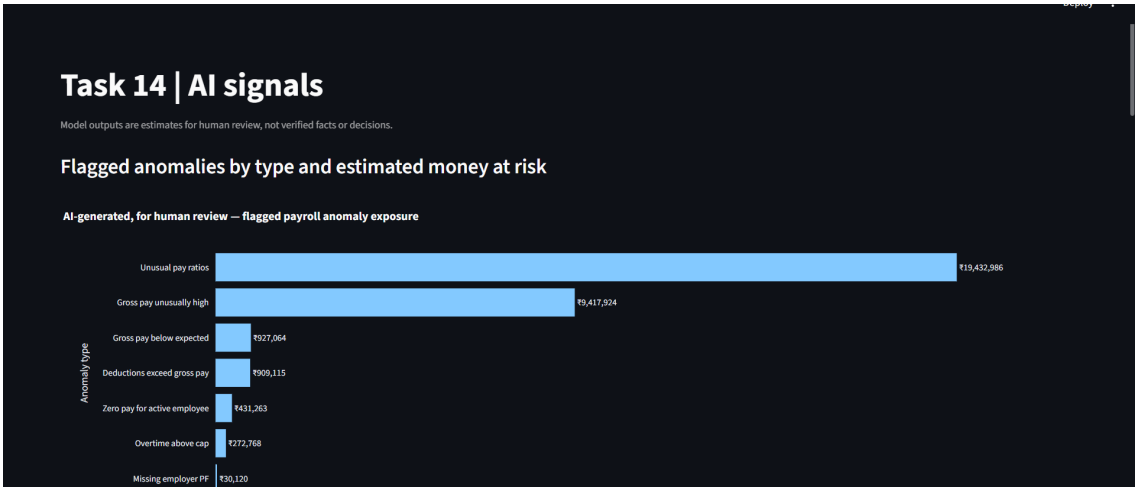


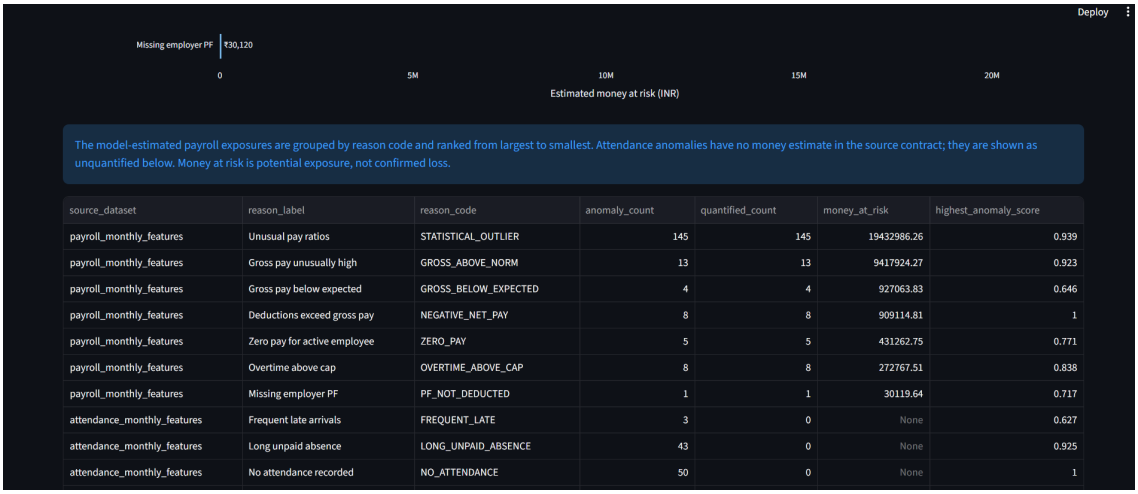
department_id	department_name	month	headcount_end	employer_cost	overtime_pay	overtime_hours	attendance_rate	approved_leave_days	le
D01	Engineering	2026-09-01 00:00:00	167	21001404.24	0	0	0.962	82	
D02	Product & Design	2026-09-01 00:00:00	26	4176418.79	0	0	0.953	10	
D03	Data & AI	2026-09-01 00:00:00	52	6424889.62	0	0	0.944	40	
D04	Sales	2026-09-01 00:00:00	46	11159004.22	0	0	0.975	12	
D05	Marketing	2026-09-01 00:00:00	13	1381169.72	0	0	0.962	7	
D06	Customer Support	2026-09-01 00:00:00	59	2920847.37	71516.6	261.5	0.924	64.163	
D07	Human Resources	2026-09-01 00:00:00	20	1635278.81	29226.82	77	0.958	13	
D08	Finance & Accounts	2026-09-01 00:00:00	17	2424967.77	30566.35	87	0.889	32.417	
D09	Operations & IT	2026-09-01 00:00:00	29	2327018.88	61686.66	174	0.959	15	
D10	Legal & Compliance	2026-09-01 00:00:00	11	1563384.45	0	0	0.933	11	



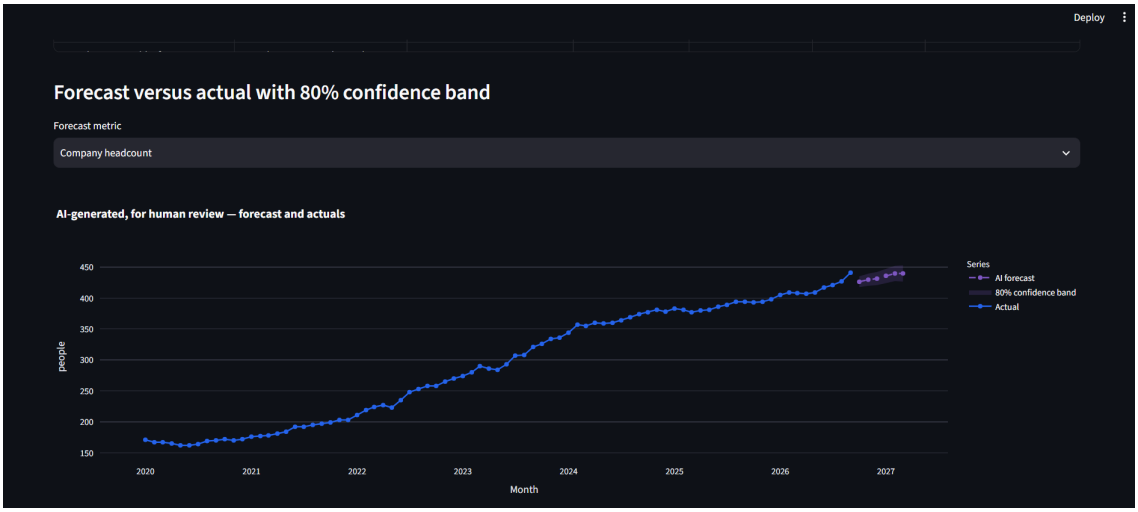
department_id	department_name	month	metric_name	actual_value	reference_value	difference	tolerance	result
empty								

Task 14. AI signals and forecasts

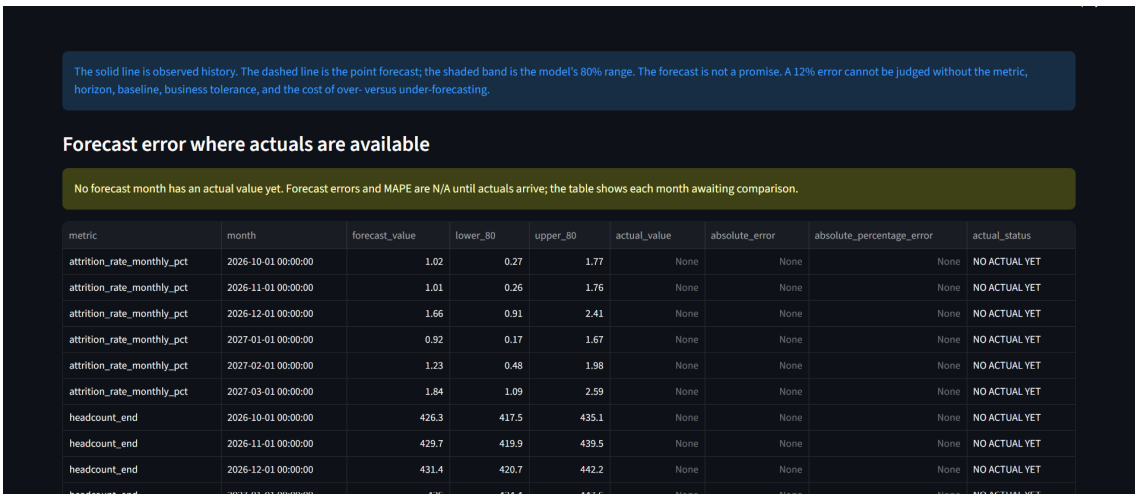




source_dataset	reason_label	reason_code	anomaly_count	quantified_count	money_at_risk	highest_anomaly_score
payroll_monthly_features	Unusual pay ratios	STATISTICAL_OUTLIER	145	145	19432986.26	0.939
payroll_monthly_features	Gross pay unusually high	GROSS_ABOVE_NORM	13	13	9417924.27	0.923
payroll_monthly_features	Gross pay below expected	GROSS_BELOW_EXPECTED	4	4	927063.83	0.646
payroll_monthly_features	Deductions exceed gross pay	NEGATIVE_NET_PAY	8	8	909114.81	1
payroll_monthly_features	Zero pay for active employee	ZERO_PAY	5	5	431262.75	0.771
payroll_monthly_features	Overtime above cap	OVERTIME_ABOVE_CAP	8	8	272767.51	0.838
payroll_monthly_features	Missing employer PF	PF_NOT_DEDUCTED	1	1	30119.64	0.717
attendance_monthly_features	Frequent late arrivals	FREQUENT_LATE	3	0	None	0.627
attendance_monthly_features	Long unpaid absence	LONG_UNPAID_ABSENCE	43	0	None	0.925
attendance_monthly_features	No attendance recorded	NO_ATTENDANCE	50	0	None	1



Task 15. Executive company-health scorecard



The solid line is observed history. The dashed line is the point forecast; the shaded band is the model's 80% range. The forecast is not a promise. A 12% error cannot be judged without the metric, horizon, baseline, business tolerance, and the cost of over- versus under-forecasting.

Forecast error where actuals are available

No forecast month has an actual value yet. Forecast errors and MAPE are N/A until actuals arrive; the table shows each month awaiting comparison.

metric	month	forecast_value	lower_80	upper_80	actual_value	absolute_error	absolute_percentage_error	actual_status
attrition_rate_monthly_pct	2026-10-01 00:00:00	1.02	0.27	1.77	None	None	None	NO ACTUAL YET
attrition_rate_monthly_pct	2026-11-01 00:00:00	1.01	0.26	1.76	None	None	None	NO ACTUAL YET
attrition_rate_monthly_pct	2026-12-01 00:00:00	1.66	0.91	2.41	None	None	None	NO ACTUAL YET
attrition_rate_monthly_pct	2027-01-01 00:00:00	0.92	0.17	1.67	None	None	None	NO ACTUAL YET
attrition_rate_monthly_pct	2027-02-01 00:00:00	1.23	0.48	1.98	None	None	None	NO ACTUAL YET
attrition_rate_monthly_pct	2027-03-01 00:00:00	1.84	1.09	2.59	None	None	None	NO ACTUAL YET
headcount_end	2026-10-01 00:00:00	426.3	417.5	435.1	None	None	None	NO ACTUAL YET
headcount_end	2026-11-01 00:00:00	429.7	419.9	439.5	None	None	None	NO ACTUAL YET
headcount_end	2026-12-01 00:00:00	431.4	420.7	442.2	None	None	None	NO ACTUAL YET

